MILITARY SPECIFICATIONS

RETICLE PROJECTOR, XM14

1. SCOPE

1.1 This specification covers one type of electro-mechanical unit known as Reticle Projector XM14 used in conjunction with electronic computer fire control systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bid form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

UU-T-106	Tape, Masking; Paper (Pressure-Sensitive)
UU-T-111	Tape, Paper Gummed (Sealing and Securing)
PPP-P-291	Fiberboard, Corrugated, Single-Face, (Flexible)
PPP-B-585	Boxes; Wood, Wirebound
PPP-B-601	Boxes; Wood, Cleated Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Boxes; Fiber
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

Military

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-130	Barrier Material, Paper, Noncorrosive
MIL-D-3464	Desiccants, Activated, Bagged Packaging use and Static Dehumidification
MIL-C-7769	Cushioning Material, Bound
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-13988	Paper, Lens Tissue, Antitarnish, Wrapping
MIL-F-45133	Fire Control Components and Subassemblies

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Components Parts
MIL-STD-20003	Indicator - Humidity, Card, Three Spot Impregnated Areas

DRAWINGS

Ordnance Corps

F10515955	Reticle Projector XM14
LSQAP10515955	Supplementary Quality Assurance Provisions List

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model. - At the beginning of production one reticle projector, representative of production shall be made available to the Government to determine if the reticle projector meets all requirements of this specification.

3.2 Materials. - Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Design and construction. - The reticle projector shall be manufactured in accordance with Drawing F10515955, Reticle Projector, XM14 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.3.1 Compatibility. - The design and performance characteristics of the reticle projector shall be such that its use does not adversely affect the capabilities of the electronic ballistic computer.

3.4 General specification. - The reticle projector shall meet the following requirements of Specification MIL-F-13926 and MIL-F-45133:

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3.4.1 MIL-F-13926

(a) General responsibilities of contractor
(b) Interchangeability (with exception to "Special Accuracy.")
(c) Lubrication
(d) Electrical
(e) Identification of parts, assemblies and complete items
(f) Workmanship

3.4.2 MIL-F-45133

(a) Tolerances
(b) Functional or performance requirements
(c) Incompatability of requirements

3.5 Performance

3.5.1 Mechanical orientation. - The reticle projector shall be energized to provide reticle illumination. The bottom surface of Key B10515975 (in Housing D1051903) shall be horizontal and the mounting surfaces above and below the key shall be in a vertical plane with the line of sight through the boresight cross perpendicular to the mounting surfaces.

3.5.2 Electrical conditions. - The complete circuitry shall be energized in accordance with Schematic Drawing D10515973 with a 400 cycle center tapped reference voltage representing plus or minus 60.075 mils applied to pins K and J and the reference voltage of 0.01 percent accuracy or better applied to pin L. The potentiometer shall have a load of 250,000 (250 K) ohms plus or minus 0.1 percent between the wiper arm and center tap.

3.5.3 Boresight knobs.

3.5.3.1 Boresight knob travel. - With the reticle projector meeting the requirements of 3.5.1 and 3.5.2, the elevation boresight knob shall be capable of moving the reticle at least 8.0 mils in elevation and 12.0 mils in depression; rotation of the deflection knob shall be capable of translating the reticle at least 10.0 mils to the right and 10.0 mils to the left as read on the reticle of an auxiliary telescope.

3.5.3.2 Backlash. - Backlash in the elevation and deflection boresight knobs shall not exceed 0.1 mils and reticle shift due to movement of either knob shall not exceed 0.2 mil in a perpendicular direction.

3.5.3.3 Torque. - With a force of not less than 1 pound nor more than 3 pounds to disengage the deflection and elevation boresight knobs from their secured position, the running torque for rotation of the knobs shall not

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exceed 6 pound-inches nor be less than 2 pound-inches at room temperature, 60 to 90 degrees Fahrenheit (°F.), and the value recorded at room temperature shall not change by more than 1 pound inch at the extreme temperatures of 3.7.1.

3.5.4 Accuracy. - With a reticle projector positioned and energized in accordance with 3.5.1 and 3.5.2, the reticle movement shall meet the tolerances specified in table 1, when the potentiometer arm is positioned by the applied voltage indicated.

TABLE I

Voltage Between Pins L & M	Potentiometer Rotation (Degrees) (Info. only)	Reticle Movement (Mils)	Tol. (Mils)
.4	36.0	1.00	±0.2
2.00	180.0	5.00	±0.2
4.00	360.0	10.00	±0.5
6.00	540.0	15.00	±1.0
8.00	720.0	20.00	±1.0

3.5.5 Slip clutch. - The slip clutch shall prevent the motor from stalling when a minimum load of 2.00 ounce inches ±10 percent is applied.

3.5.6 Reticle.

3.5.6.1 Reticle accuracy. - The total angular subtense at the extremeties of the series of horizontal reticle lines through the boresight circle shall be 29.5 mils plus or minus 1.0 mil.

3.5.6.2 Reticle plumb. - The vertical reticle line shall be vertical within 1 degree when positioned in accordance with 3.5.1.

3.5.6.3 Parallax. Parallax between the image of the projected reticle and the reticle of an auxiliary telescope adjusted for zero parallax at infinity shall not exceed 0.1 mil.

3.6 Electrical.

3.6.1 Continuity. - The electrical circuits shall be tested for continuity and shall conform to the Schematic Drawing, D10515973.

3.6.1.1 Dimmer control. - The dimmer control rheostat knob shall be capable of varying intensity of the reticle image from high to low intensity when turning in a clockwise direction. There shall be an "OFF" position on the rheostat at the full clockwise direction.

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3.6.2 Insulation.

3.6.2.1 Dielectric strength. - There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength when the insulated area of the reticle projector are subjected to an alternating current (a-c) root mean square (rms) potential of 1000 volts twice the rated voltage at a frequency of 60 cycles per second for a period of not more than 30 seconds nor less than five seconds.

3.6.2.2 Insulation resistance. - Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (d-c) applied.

3.7 Environmental.

3.7.1 Operating temperatures. - The reticle projector shall meet all the performance requirements of this specification while exposed and thermally stabilized at ambient temperatures of plus 150 degrees Fahrenheit and minus 40 degrees Fahrenheit (plus 65 degrees Centigrade and minus 40 degrees Centigrade).

3.7.2 Storage temperatures. - The reticle projector shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature (plus 60 to 90 degrees Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit (plus 71 degrees and minus 62 degrees Centigrade).

3.7.3 Vibration. - With the reticle projector mounted in its normal operating position, it shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration there shall be no evidence of loose parts or damage and the reticle projector shall meet all the following performance requirements of this specification.

3.7.3.1 Cleanliness. - Subsequent to vibration, the reticle projector shall meet the cleanliness requirements of MIL-O-13830, and there shall be no more than one particle of dirt visible on the reticle and the area of the particle shall not be in excess of the area equivalent to a size ten dig.

3.7.4 Humidity. - The reticle projector shall be exposed to one complete humidity cycle as shown on Figure 1 and shall subsequently meet all the performance requirements of this specification.

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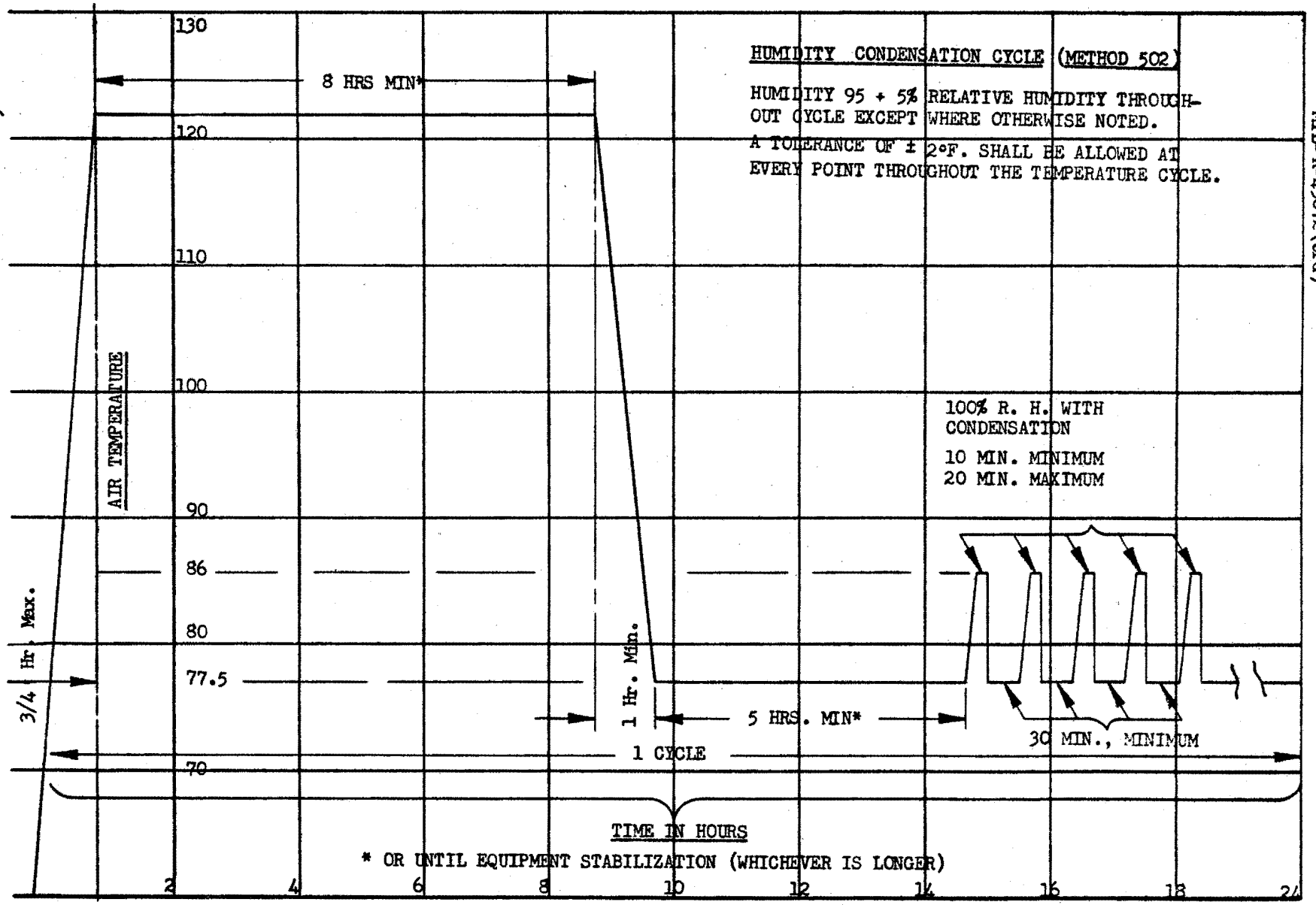


FIGURE I

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3.7.5 Sealing. - The reticle projector shall show no evidence of leakage in excess of 0.1 psi when subjected to an internal pressure of 5.0 psi (gage pressure) for a period of 5 minutes. The interior atmosphere of the unit shall consist of dry nitrogen having a dew point of at least 25 degrees Fahrenheit at atmospheric pressure.

3.8 Identification marking. - Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering all components and assemblies of this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when as a minimum it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

The contractor will not be restricted to the inspection station or to the method of inspection listed in the written procedure provided an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of applicable drawings and specifications shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification and the other documents referenced herein.

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4.1.3 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of:

(a) Surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied.

(b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.1.4 Inspection Equipment. - Contractor inspection referred to in par. 4.1.1 shall be performed with such inspection equipment consisting of special testing equipment, measuring instruments and gages as necessary for conformance to this specification. This equipment and maintenance thereof shall be a part of contractor's quality assurance system of par. 4.1.2.

4.1.4.1 The Government inspector will determine that the contractor has available, and utilizes correctly such equipment referred to in the above paragraph.

4.1.4.2 The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, a set of mastergages standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. Calibration equipment will be periodically checked and certified by authorized Government personnel.

4.1.4.3 Accuracy Requirements. - Inspection equipment used in performance of this specification shall be capable of measuring within an accuracy of 10 percent of the specified product requirement tolerance. In cases where this requirement is seemingly impractical, the Government shall be expediously advised in order that necessary appropriate action can be taken.

4.2 Initial inspection and qualification inspection.

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4.2.1 Initial inspection. - The production projector shall be submitted for inspection and tests as specified in 3.1. Inspection shall be performed at the place of manufacture. Failure of the sample to comply with any of the requirements specified shall be cause for refusal of the Government to conduct a retest until corrective measures, satisfactory to the Government, have been taken.

4.2.2 There are no Qualification Inspection Requirements.

4.3 Inspection provisions.

4.3.1 Lot formation. - Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105. Defective product shall also be processed as specified in that specification.

4.3.2 Examination inspection.

4.3.2.1 General examination. - Quality Assurance Provisions specified in MIL-F-13926, MIL-F-45133 and MIL-P-116 shall apply, where applicable, and are not in conflict with the provisions specified herein.

4.3.2.2 Components and sub-assemblies. - Classification of defects and acceptable quality levels contained on SQAP's listed on LSQAP #10515955 shall represent the contractor's minimum inspection provision requirements for the item and shall be the Government's basis for acceptance or rejection by lot. The Government reserves the right to inspect for all the requirements of all applicable drawings and specifications and to reject individual non-conforming items.

4.3.2.3 Complete assembled item. - The classification of defects and acceptable quality levels listed in the following tables shall constitute the contractor's minimum examination inspection requirements for the end item and will be the Government's basis for acceptance or rejection by lot. The Government reserves the right to inspect for the requirements of all applicable drawings and specifications and to reject individual non-conforming items.

TABLE II CLASSIFICATION OF DEFECTS

EXAMINATION INSPECTION

Use Inspection Level II and Sampling Plan Table IVA of Standard MIL-STD-105

AQL value is expressed in percent defective

CHARACTERISTIC TO BE INSPECTED (60°F. to 90°F.)

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MAJOR: AQL 1.5%	Req't Par.	Q.A. Par.	Method of Inspection
Performance			
101. Mechanical orientation	3.5.1	4.4.1.1	Special Test Equipment (STE)
102. Electrical conditions	3.5.2	4.4.1.2	STE
103. Boresight knob travel	3.5.3.1	4.4.1.3	STE
104. Boresight knob backlash	3.5.3.2	4.4.1.4	STE
105. Boresight knob torque	3.5.3.3	4.4.1.5	STE
106. Projector accuracy	3.5.4	4.4.1.6	STE
107. Slip clutch	3.5.5	4.4.1.7	STE
108. Reticle accuracy	3.5.6.1	4.4.1.8	STE
109. Reticle plumb	3.5.6.2	4.4.1.9	STE
110. Parallax	3.5.6.3	4.4.1.10	STE
111. Lack of continuity	3.6.1	4.4.1.11	Measuring Equip. (ME)
112. Dielectric strength failure	3.6.2.1	4.4.1.12	ME
113. Insulation resistance failure	3.6.2.2	4.4.1.13	ME
MAJOR: 2.5			
Packaging, Packing and Marking	5	4.4.3	
114. Cleanliness of projector, hardware, and packing materials (Levels A,C)	5.1.1.1 5.1.1.2		Visual
115. Window not covered and taped as specified (Levels A,C)	5.1.1.3.1 5.1.2.2		Visual
116. Mounting hardware not wrapped and secured as spec. (Levels A,C)	5.1.1.3.2 5.1.2.2		Visual
117. Projector, power unit and harness not wrapped, overwrapped and taped as specified	5.1.1.3.3 5.1.2.2		Visual
118. Improper cushioning and immobilization in inner container (Levels A,C)	5.1.1.3.4 5.1.2.2		Visual
119. Desiccant not as specified (Level A)	5.1.1.3.5		Visual
120. Projector in inner container improperly cushioned, immobilized, and closed (Levels A,C)	5.1.1.3.6 5.1.2.2		Visual

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MAJOR: 2.5	Req't Par.	Q.A. Par.	Method of Inspection
Packaging, Packing and Marking			
121. Waterproof bag and humidity indicator not as specified (Level A)	5.1.1.3.7		Visual
122. Outer container not as specified (Level A)	5.1.1.3.8		Visual
123. Markings, illegible, incorrect, not conforming to spec. (Level A, B, C)	5.3		Visual MIL-STD-129 MIL-P-116
MINOR: AQL 4.0			
201. Shipping container not conforming to spec. (Levels A, B, C)	5.2.1 5.2.2 5.2.3		ME PPP-B-601 PPP-B-621 PPP-B-585 PPP-B-636
202. Strapping improperly applied (Level A, B, C)	5.2.4		Visual
203. Workmanship (Level A, B, C)			Visual

4.3.3 Inspection testing.

4.3.3.1 Control of materials. - Materials from which the item is manufactured or fabricated shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or sub-contractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.3.3.2.- The characteristics to be tested, and acceptable quality levels listed in the following tables shall constitute the contractor's minimum inspection requirements for inspection testing and will be the basis for Government acceptance and rejection by lot. The Government reserves the right to inspect for the requirements of all drawings and specifications and to reject individual non-conforming items:

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TABLE III INSPECTION TESTING

Use Inspection Level L7 and Sampling Plan Table IVA of MIL-STD-105

AQL value is expressed in percent defective

CHARACTERISTIC TO BE TESTED AQL 6.5	Req't Par.	Q.A. Par.	Test Method or Procedure
1. Operating temp. failure	3.7.1	4.4.2.1	STE
2. Storage temp. failure	3.7.2	4.4.2.2	STE
3. Vibration failure	3.7.3	4.4.2.3	STE
4. Post vibration cleanliness	3.7.3.1	4.4.2.4	Visual
5. Humidity	3.7.4	4.4.2.5	STE
6. Sealing	3.7.5	4.4.2.6	Visual

4.3.4 Inspection equipment.

4.3.4.1 Government equipment. - If an index to inspection equipment list is available and specified in the invitation to bid, or by the contracting officer, inspection equipment referenced on the index will be utilized by the Government as standards for determining the adequacy of the contractor's inspection equipment in the inspection of product requirement characteristics listed on the individual Inspection Equipment Lists. Ordnance drawings of such equipment will be made available to the bidder for information purposes upon request.

4.3.4.2 Special test equipment (STE).

4.3.4.2.1 Operating and storage temperature. - The special test equipment used for the operating and storage temperature tests shall be capable of maintaining the extreme temperatures required within a tolerance of plus or minus 2° Centigrade until thermal stabilization is reached.

4.3.4.2.2 Humidity. - The special test equipment used for the humidity test shall be capable of maintaining the relative humidity within plus or minus 5% of the relative humidity reading.

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4.3.4.2.3 Vibration. - The special test equipment shall be capable of producing the vibration frequencies and cycling required and the frequency measuring instruments used with the device special testing shall have an accuracy of plus or minus 2 percent.

4.3.4.2.4 Performance. - The special test equipment utilized to check the performance requirements shall mount the reticle projector in its normal operating position with the bottom surface of the key horizontal within 10 seconds and the line of sight perpendicular to the mounting surface. The device shall provide a means of energizing the complete circuitry in accordance with Schematic Drawing D1051973. The potentiometer shall have a load of 250K ohms plus or minus 0.1 percent between the wiper arm and the center tap. The device shall contain an auxiliary telescope adjusted for zero (0) parallax at infinity, and will permit auto collimation of the auxiliary telescope with the projected image of the reticle projector. The auxiliary telescope reticle shall be clearly defined and the lines shall not exceed .0010 in width. Angular tolerance shall not exceed five seconds (0°0'5"). The auxiliary telescope reticle pattern shall permit testing the following requirements; reticle accuracy, reticle plumb, parallax and reticle movement values listed in Table I.

4.3.4.2.5 Torque. - The equipment shall have a minimum range of from zero (0) to eight (8) inch pounds, the accuracy of which shall be plus or minus 1/4 inch pound throughout the entire zone of excursion in both clockwise and counterclockwise direction, equipped with suitable adapters to accommodate testing the boresight knobs.

NOTE: All special set-up and calibration equipment, necessary for certification of equipment accuracy shall be furnished and will become a part of the special test equipment.

4.4 Test methods and procedures.

4.4.1 Performance.

4.4.1.1 Mechanical orientation. - The reticle projector shall be mounted on special test equipment and oriented in accordance with the requirements specified.

4.4.1.2 Electrical conditions. - The electrical conditions shall be checked for compliance with requirements specified.

4.4.1.3 Boresight knob travel. - With the reticle projector mounted on special test equipment, oriented and energized, manipulation of the boresight knobs shall translate the reticle laterally and vertically in compliance with the requirements.

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4.4.1.4 Backlash. - Backlash in the elevation and deflection boresight knobs and reticle shift due to boresight knob movements, shall not exceed the allowable tolerances.

4.4.1.5 Torque. - The force required to disengage the boresight knobs from their secured position and the running torque required to rotate the knobs shall comply with the requirements.

4.4.1.6 Accuracy. - With the reticle projector mounted on special test equipment, oriented, energized, the reticle movement shall be within the tolerances listed in the table I when the indicated voltages are applied to the potentiometer.

4.4.1.7 Slip clutch. - The function of the slip clutch shall be checked for compliance with the minimum load requirements.

4.4.1.8 Reticle accuracy. - The accuracy of the reticle pattern shall comply with the requirements when measured with suitable special test equipment.

4.4.1.9 Reticle plumb. - With the reticle projector positioned on a device, special testing, the vertical reticle line shall be plumb within the required tolerance.

4.4.1.10 Parallax. - Parallax between the image of the projector reticle and the reticle of an auxiliary telescope adjusted for zero parallax at infinity shall be within the required tolerance.

4.4.1.11 Continuity. - A continuity check of the reticle projector circuits shall be made using standard measuring equipment to determine conformance with the requirements specified.

4.4.1.12 Dielectric strength. - The reticle projector shall be tested in accordance with method 301 of MIL-STD-202, and shall meet the requirements specified.

4.4.1.13 Insulation resistance. - Immediately after satisfactorily withstanding the test for dielectric strength, the reticle projector shall be given the insulation resistance test in accordance with method 302, test condition B, of MIL-STD-202, and shall meet the requirements specified.

4.4.2 Environmental.

4.4.2.1 Operating temperatures. - When thermal equilibrium of the reticle projector is reached at plus 150°F and minus 40°F, respectively, the unit shall be subjected to the test specified in 4.4.1.1 to 4.4.1.13 and shall meet the requirements specified.

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4.4.2.2 Storage temperature. - The test shall be conducted in accordance with the appropriate portions of procedure II of MIL-F-13926 to the conditions specified. Upon completion of the specified exposure, the unit shall be inspected for damage and then subjected to the test specified in 4.4.1.1 to 4.4.1.13 and shall meet the requirements specified.

4.4.2.3 Vibration. - The reticle projector shall be mounted on a device, special testing and vibrated at the frequency specified for the time period indicated.

4.4.2.4 Cleanliness. - Subsequent to vibration the reticle projector shall be examined for compliance with the cleanliness requirements.

4.4.2.5 Humidity. - The reticle projector to be tested shall be placed in the test chamber in its normal operating position. The test shall be conducted in accordance with applicable portions of procedure I of MIL-F-13926 to the conditions specified in 3.5.4. Upon completion of the specified humidity cycle, the unit shall be visually inspected for damage and then subjected to the test specified in 4.4.1.1 to 4.4.1.13 and shall meet the requirements specified.

4.4.2.6 Sealing. - The reticle projector shall be tested for compliance with the requirements specified.

4.4.3 Packaging, packing and marking tests. - Tests for the determination of cleanliness, wrapping, securing, cushioning of the reticle projector shall be performed to the extent necessary to assure proper product compliance with requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C as specified, see (6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - Cleaning process C-1 of Specification MIL-P-116 shall be followed to clean all exterior surfaces of the reticle projector, power unit, harness and mounting hardware.

5.1.1.2 Drying. - Immediately after cleaning, the items shall be dried thoroughly in accordance with any applicable procedure specified in Specification MIL-P-116 that will not damage the items.

5.1.1.3 Unit packaging. - Packaging shall be as specified herein, which is in conformance with method IIb of Specification MIL-P-116.

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5.1.1.3.1 Covering the window. - The window shall be covered with lens tissue paper conforming to Specification MIL-P-13988, which shall be secured in place with pressure sensitive tape conforming to type I of Specification UU-T-106.

5.1.1.3.2 Wrapping and securing mounting hardware. - The mounting hardware shall be placed together and wrapped with barrier material conforming to type II of Specification MIL-B-130. The wrap shall be secured in place with pressure sensitive tape conforming to type I of Specification UU-T-106. The wrapped hardware shall be secured to the reticle projector with pressure sensitive tape conforming to type I of Specification UU-T-106.

5.1.1.3.3 Wrapping the reticle projector, power unit and harness. - The items shall be wrapped separately with barrier material conforming to type II of Specification MIL-B-130 and overwrapped with flexible single face corrugated fiberboard conforming to type I of Specification PPP-P-291. The wrap shall be secured with gummed tape conforming to class 2 of Specification UU-T-111.

5.1.1.3.4 Inner container. - The wrapped items shall be cushioned and immobilized in a fiberboard box conforming to type I, class I, of Specification PPP-B-636.

5.1.1.3.5 Desiccant. - The proper amount of dry desiccant, determined by formula I of Specification MIL-P-116, shall conform to Specification MIL-D-3464 and placed within the inner container.

5.1.1.3.6 Cushioning, immobilizing and closure of inner container. - The interior surfaces of the inner container shall be lined with 2-inch thick fiber bound cushioning material conforming to type III or IV of Specification MIL-C-7769. The items shall fit snugly in the bound cushioning material. The inner container shall be closed with 2-inch width gummed tape conforming to class 2 of Specification UU-T-111. Apply one strip on the top and bottom seam, extend 3" on to each end panel.

5.1.1.3.7 Water-vaporproof bag and humidity indicator. - The inner container shall be enclosed in a water-vaporproof bag conforming to type I, class e of Specification MIL-B-117. A humidity indicator card conforming to Military Standard MS-20003 shall be included within the bag. The indicator shall be located at the opening of the bag where it can readily be seen for subsequent inspections. The opening of the bag shall be heat sealed in such a manner that the bag can be opened and resealed at least twice, and pass the heat seal and quick leak tests of Specification MIL-P-116. Air shall be exhausted from the bag prior to final closure.

5.1.1.3.8 Outer container. - The sealed package shall be placed in a close fitting fiberboard box conforming to the requirements for type I, class 2 of Specification PPP-B-636. The fiberboard box shall be closed with pressure sensitive tape conforming to PPP-T-76 in accordance with the appendix of Specification PPP-B-636.

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5.1.2 Level C.

5.1.2.1 Cleaning and drying. - Cleaning and drying shall be as specified in 5.1.1.1 and 5.1.1.2.

5.1.2.2 Unit packaging. - Unit packaging shall be as specified in 5.1.1.3.1 to 5.1.1.3.6 inclusive, except as follows, which is in conformance with method III of Specification MIL-P-116.

(a) The desiccant, specified in 5.1.1.3.5, shall be omitted.

(b) The inner container shall be closed as specified in the appendix of Specification PPP-B-636.

5.2 Packing. - Packing shall be level A, B, or C as specified, see 6.2.

5.2.1 Level A. - The quantity of unit packaged reticle projectors indicated in 6.2 shall be immobilized in a wood-cleated plywood box or nailed wood box conforming to the requirements of Specification PPP-B-601, overseas type or PPP-B-621, class 2, respectively.

5.2.2 Level B. - The quantity of unit packaged reticle projectors indicated in 6.2 shall be immobilized in a wood-cleated plywood box, nailed wood box, or wirebound box conforming to the requirements of Specifications PPP-B-601, domestic type, PPP-B-621, class I, or PPP-B-585, class 1, respectively.

5.2.3 Level C. - The quantity of unit packaged reticle projectors indicated in 6. shall be immobilized in a wood-cleated plywood box, nailed wood box, wirebound box, or fiberboard box conforming to the requirements of Specifications PPP-B-601, domestic type, PPP-B-621, class I, PPP-B-585, class I, or PPP-B-636, type I or II, class I, respectively.

5.2.4 Strapping. - Nailed wood boxes and plywood boxes shall be strapped as specified in the applicable box specification. Straps shall not be stapled on a level B or C shipping container.

5.3 Marking. - In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129 and Specification MIL-P-116. Fragile labels shall be affixed to the shipping container in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - The Reticle Projector, XM14 is an electro-mechanical instrument which receives an input signal from the servo-amplifier of the computer unit, a component of the electronic fire control system. This

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electrical signal is received by the motor-tachometer which results in deflection movement of the reticle in the reticle projector and is nulled by a potentiometer. The deflection value represents azimuth correction. The reticle is projected into the 8-power system of the gunner's periscope for final lay on a target. The effective focal length (EFL) of the reticle is 8.819 inches.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of an applicable level of preservation, packaging and packing.

(c) Applicable stock number.

(d) Provisions for production model including date of delivery, see 3.1.

6.3 Design information. - Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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